1. Develop prompt templates to support the core functionality of your project.

We developed standardized prompt templates to support the core functionality of the system: automated, executive-facing explanation of KPI anomalies grounded in precomputed statistical analysis.

Prompts were designed to (a) minimize hallucination, (b) clearly separate evidence from interpretation, and (c) produce outputs suitable for executives.

Core Prompt Template

Purpose: Generate a structured root-cause explanation from a detected anomaly and statistical findings.

Template Structure:

- System role definition (analytics assistant)
- Explicit domain context (e-commerce KPIs)
- Provided dataset columns
- Detected anomaly description
- Precomputed statistical findings
- Constraints against unsupported causal claims
- Required output format

2. Experiment with different prompting techniques, including:

Instruction-based prompting

You are a KPI root-cause analysis assistant for an e-commerce business.
You must ONLY use the facts explicitly provided below.
Do NOT invent metrics, variables, or causes.
If information is missing, state the uncertainty and propose analytical checks.

Dataset columns: date, revenue, orders, traffic, conversion_rate, avg_order_value, region, device_type, marketing_spend, product_category

Detected anomaly: Revenue dropped 15% starting April 20, 2024 ($p < 0.01$).

Statistical findings (precomputed):

- Primary driver: Mobile conversion_rate dropped 22% ($r = 0.71$, $p < 0.01$)
- Secondary factor: marketing_spend in the West region decreased 35%
- Ruled out: avg_order_value (no significant change)
- Ruled out: product_category mix (stable)

Constraints:

- Base all claims strictly on the findings above.
- Clearly separate EVIDENCE from HYPOTHESES.
- Do not introduce unsupported causes.

RESPONSE FORMAT:
1) Executive summary (2–3 sentences)
2) Key evidence (bullets; use only provided numbers)
3) Hypotheses (bullets; clearly labeled)
4) Recommended next analyses (bullets)

Few-shot prompting

You are a KPI root-cause analysis assistant. Use ONLY provided data.

- Anomaly: Customer churn increased 10% in Q2.
- Findings: Support ticket volume increased 25%; NPS declined.

Response:

- Executive summary:
- Churn increased primarily alongside rising support issues.

Key evidence:

- Ticket volume +25%
- NPS declined

Hypotheses:

- Service delays may be affecting retention.

Next analyses:

- Analyze ticket resolution times.

Dataset columns:
date, revenue, orders, traffic, conversion_rate, avg_order_value, region, device_type, marketing_spend, product_category

Anomaly: Revenue dropped 15% starting April 20, 2024.

Findings:

- Mobile conversion_rate dropped 22% ($r = 0.71$, $p < 0.01$)
- Marketing spend in the West region decreased 35%
- Avg_order_value and product_category mix unchanged

Generate an executive-facing explanation using the same structure as the example.

Chain-of-thought prompting
You are a data analyst.

Dataset columns: date, revenue, orders, traffic, conversion_rate, avg_order_value, region, device_type, marketing_spend, product_category

Anomaly: Revenue dropped 15% starting April 20, 2024.

Statistical findings (precomputed):
- Primary driver: Mobile conversion_rate dropped 22% ($r = 0.71$, $p < 0.01$)
- Secondary factor: marketing_spend in the West region decreased 35%
- Ruled out: avg_order_value (no significant change)
- Ruled out: product_category mix (stable)

Step by step:
1. Reason through which variables could explain the revenue drop.
2. Identify which findings are strongest and why.
3. Distinguish evidence from speculation.
4. Produce a final executive summary.

3. Create a test suite that includes representative and edge-case inputs.
Presentative Cases
- single-driver anomaly
- multi–factor anomaly
- ruled-out factors

Edge Cases
- conflicting signals, e.g., multiple KPIs change in opposing directions
- sparse findings, e.g., limited statistical evidence provided
- high ambiguity, e.g., correlations without clear temporal alignment
- hallucination traps, e.g., prompts containing tempting but unsupported causal narratives

4. Evaluate prompt outputs for quality and consistency, and iterate on prompt designs based on the results.
SYSTEM:
- You are a KPI root-cause analysis assistant for an e-commerce business.
- Your task is to generate executive-facing explanations of KPI anomalies.
- You must ONLY use the facts explicitly provided below.
- Do NOT invent metrics, variables, or causes.

- If information is missing, explicitly state the uncertainty and propose analytical checks.
- Internally reason carefully, but do NOT expose your reasoning.

USER INPUT:
Dataset context: This dataset contains e-commerce sales and marketing metrics with the following columns:
date, revenue, orders, traffic, conversion_rate, avg_order_value, region, device_type, marketing_spend, product_category

Detected anomaly: Revenue dropped 15% starting April 20, 2024 ($p < 0.01$).

Statistical findings (precomputed):
- Primary driver: Mobile conversion_rate dropped 22%, strongly correlated with revenue ($r = 0.71$, $p < 0.01$).
- Secondary factor: marketing_spend in the West region decreased 35%.
- Ruled out: avg_order_value (no statistically meaningful change).
- Ruled out: product_category mix (stable over time).

Constraints:
- Base all claims strictly on the findings above.
- Clearly separate EVIDENCE from HYPOTHESES.
- Do not introduce unsupported causes.
- Do not restate the dataset description in the response.

RESPONSE FORMAT (user can adapt):
1. Executive summary (2–3 sentences, executive-facing)
2. Key evidence (bullet points; use only provided numbers)
3. Hypotheses (bullet points; clearly labeled as hypotheses, not facts)
4. Recommended next analyses (bullet points; analytical checks only)